

Estimating Ideology and Influence in Social Media Networks: Opinions on Genome Editing in Livestock

Estimation Using R and RStan Code

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ARTICLE HISTORY

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1. Introduction

We provide supplementary materials to help readers replicate our methods applied in our paper. For a more detailed explanation of the methods applied in our paper, please also refer to [1]. We provide detailed information about the machine and software used in the analysis because the computational methodology and results may vary across machine and versions when using R and RStan [2,3]. For a deeper explanation, please refer to the [Reproducibility Chapter](#) from the [RStan website](#).

The machine used for the following analysis is:

- 2018 MacBook Pro - macOS: Monterey Version 12.4, Processor: 2.7 GHz Quad-Core Intel Core i7, SSD: 1T, Memory: 16 GB, and Graphics: Intel Iris Plus Graphics 655 1536 MB.

The software needed to replicate our analysis is:

- R - Version 4.2.2 (2022-10-31)
- RStudio - Version 2022.07.2, “Spotted Wakerobin” Release (e7373ef8, 2022-09-06) for macOS
- RStan - Version 2.26.1 on macOS. To download RStan for your machine, please visit the main [Website](#) or [GitHub](#) [2].

Please refer to the point of contact and their GitHub page for a zip file titled *GEL_Ideology_and_Influence_Tweet.zip*. This contains these files needed to replicate our analysis:

- *GEL_Ideology_and_Influence_Tweet_RScript.R* - This is the main R script the user should run to replicate our methods. All additional material will be an input into this file. The packages needed to run this file are listed at the top of the file's script. We heavily annotate this script to guide the reader through the replication process.
- *StanModel-FixedLambda-Ident.R* - This is the **RStan** File that is needed to conduct the MC-MC estimation. It contain the Jeffreys priors and the identification approach described in the paper. We do not recommend altering this file without first fully understanding how to write **Stan** code.
- *Bayes_Tweet_Functions.R* - This file contains a set of additional functions used in the analysis. It mostly contains plotting functions to help readers visualize the data and the priors used in the analysis.
- *sim_y_mat.rdata* - This is the main data file used in the analysis. It is a synthetically generated dataset, which was uniquely generated based on the model fitted to the original Twitter dataset from the main paper. This dataset was generated to ensure the proprietariness of the original Twitter data, and to help readers replicate our methods in a transparent way. In general, the dataset is simply a matrix with n rows of followers and m columns of experts with elements 1 if the follower follows the expert, 0 or . otherwise.
- *GEL_expert_profile_info.csv* - Contains the demographic information for all of the Anti-GEL and Pro-GEL influencers. The data contains their name, username (Twitter handle), account description, their assumed position on GEL, and if they are verified on Twitter.

2. Instructions and Expected Output for Sanity Checks

Follow the steps below to perform a Bayesian analysis on a simulated Twitter data stored in *sim_y_mat.rdata*.

- Create a directory or folder called Supp in the file system on your laptop or desktop PC. For this demonstration, we will assume that the full path to that folder is "C:/Users/MyAcct/Demo". Unzip the contents of the file *GEL_Ideology_and_Influence_Tweet.zip* in this directory. If a folder, possibly named GEL_Ideology_and_Influence_Tweet, containing the files is created, just move the files over one directory up.
- Run the RStudio program and open the script file *GEL_Ideology_and_Influence_Tweet_RScript.R*. To do this, click on **File** on the menu bar (top left below the RStudio logo) and click on **Open File . . .**. Use the new window to navigate through the file system to locate file's directory, then enter the file's name in the **File name:** box. Click on **Open**.

- The script file contains all the commands that produce results similar to those in the paper. We suggest running the codes in stages or batches. We include comments in the script for guidance and to indicate which commands to run in stages. To run a group of consecutive commands, highlight them and click the **Run** button on the top-right of the script window.

Below, we provide plots and tables of the expected output from running the *GEL_Ideology_and_Influence_Tweet_RScript.R* file. These plots and tables are meant to be sanity checks for readers, users, and researchers who would like to replicate our analysis or apply our methods to their own data.

2.1. The Synthetically Generated Input Data from the Twitter Data Analysis

Figure 1 is a plot of the raw synthetically generated dataset *Sim_Y_Mat.rdata*. This is a visualization of the input data used in the analysis. The columns are indexed by the $n = 46$ experts and the rows by the $m = 3,383$ followers. There are $N = m \times n = 155,618$ values of p_{ij} that will be estimated.

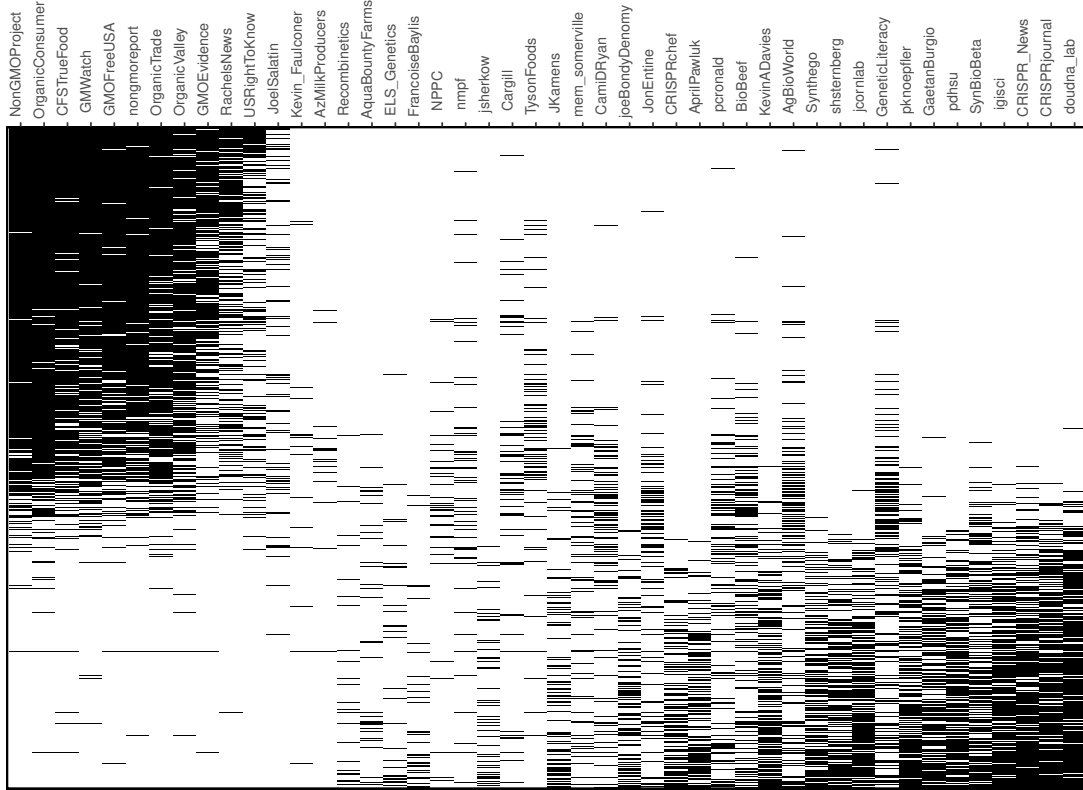


Figure 1. A heat map of the simulated connections (following relationships) between 3,383 followers and 46 GEL experts on Twitter.

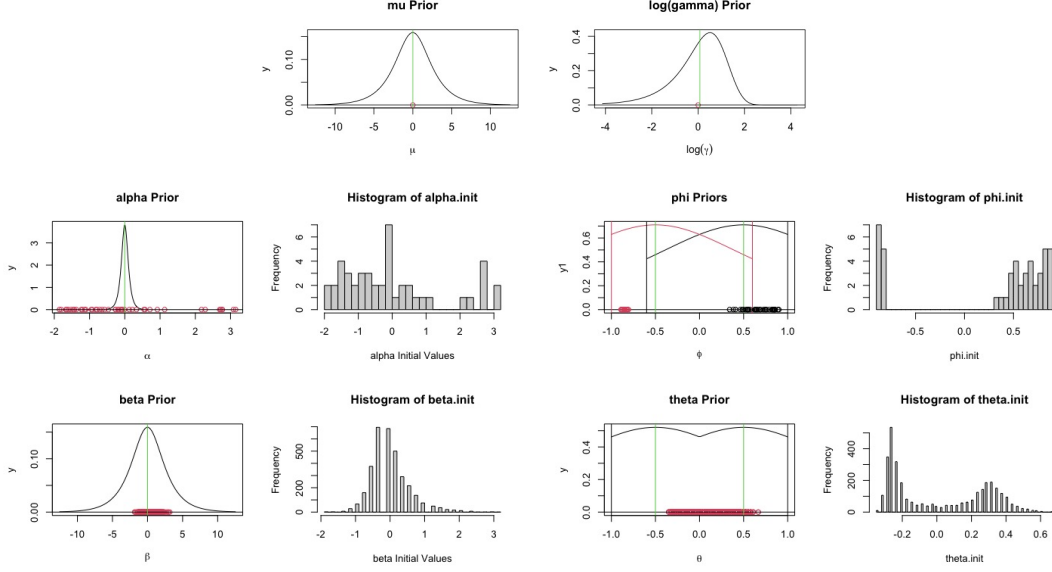


Figure 2. Jeffreys Prior Distributions and Initial Value Plots from the Estimation Code

2.2. Jeffreys Prior Distributions and Initial Value Plots

Figure 2 shows the prior distributions and initial values used in the MC-MC estimation. There are seven prior distribution plots and four histograms of initial values. The prior distributions also show the points we used for initial values indicated by small red circles. The top row of plots are the prior distributions for μ and γ , and the bottom two rows are the prior distributions and histograms of initial values for α_j , ϕ_j , β_i , and θ_i . These plots are outputs in the estimation code and they should be used as a sanity check when replicating the analysis. Additionally, each prior distribution has its own set of location and scale parameters, which we pre-define in the replication code to get these distributional shapes.

2.3. Numerical and Graphical Results

2.3.1. Estimates of Expert Popularity and Ideal Points

Figure 3 plots Bayesian estimates of popularity ($\hat{\alpha}_j$) and ideal point ($\hat{\phi}_j$) based on the posterior distribution for each expert j generated from the MC-MC estimation. Each expert has a box-plot with the “whiskers” indicating the ends of the 95% credible intervals. This figure closely resembles the results and figures from our paper, and we refer the reader to our paper and [1] for a deeper interpretation of the plot.

2.3.2. Estimates of Follower Engagement and Ideal Points and Other Model Parameters

Figures 4 (a) and (b) present histograms of the posterior means for the engagement estimates ($\hat{\beta}_i$) and ideal point estimates ($\hat{\theta}_i$) for all of the i followers, respectively. These figures closely resemble those in our paper, and we refer the reader to our paper for a deeper interpretation of the plots. We also present the summary statistics for

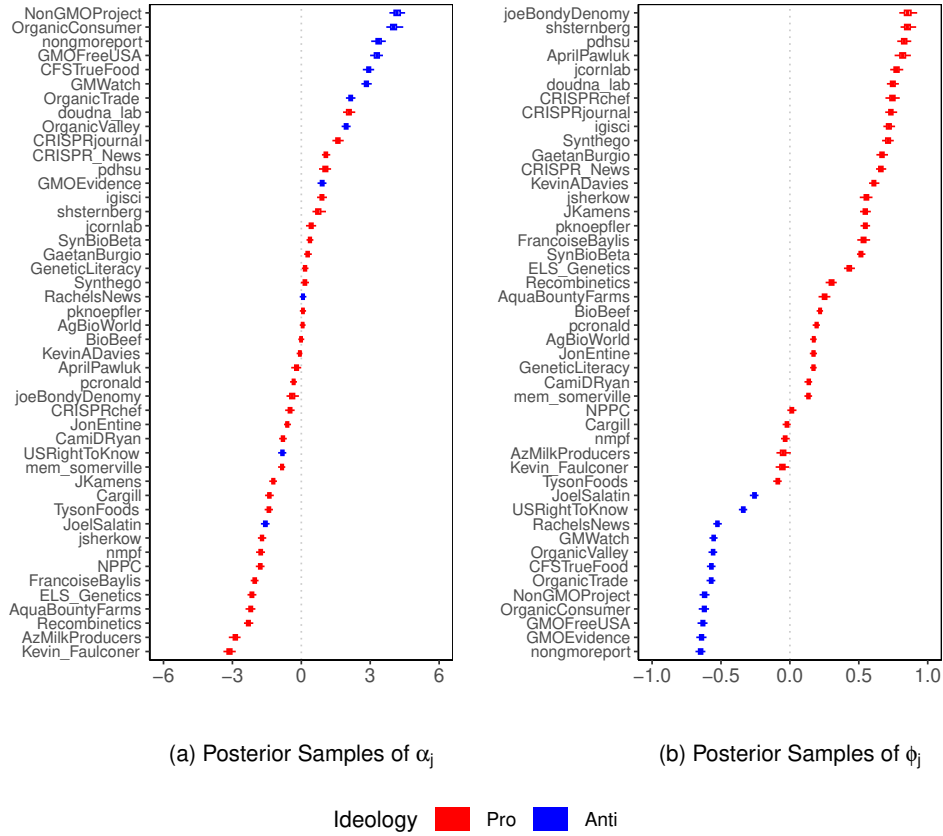


Figure 3. Posterior distributions of (a) popularity ($\hat{\alpha}_j$) and (b) ideal point ($\hat{\phi}_j$) with 95% credible intervals for all experts in the simulated Twitter data. Anti-GEL and Pro-GEL experts are colored red and blue, respectively.

the posterior distributions of engagements, ideal points, and other model parameters in Table 1. R codes for creating a simplified form of this table in LaTeX are given in the file *Bayes.Tweet.RScript.R*.

2.4. Diagnostics

2.4.1. MC-MC Diagnostics

Figure 5 presents the $\hat{R}$ values for all of the original and adjusted (identifiable) parameters that are estimated in the analysis. Note that all of the $\hat{R}$ values are below the 1.1

Table 1. Summary Statistics and 95% Credible Intervals for the Engagement $\hat{\beta}_i$ and Ideal Point $\hat{\theta}_i$ Estimates for the 3,383 Followers, and for μ and γ in the Simulated Twitter Data Analysis

Estimator	Mean	Median	SD	Lower 95% CI	Upper 95% CI
$\hat{\beta}_i$	0.039	-0.016	0.799	-1.412	1.903
$\hat{\theta}_i$	-0.056	-0.148	0.583	-0.763	0.871
$\hat{\mu}$	-8.981	-8.965	0.223	-9.427	8.572
$\hat{\gamma}$	9.615	9.606	0.228	9.158	10.068

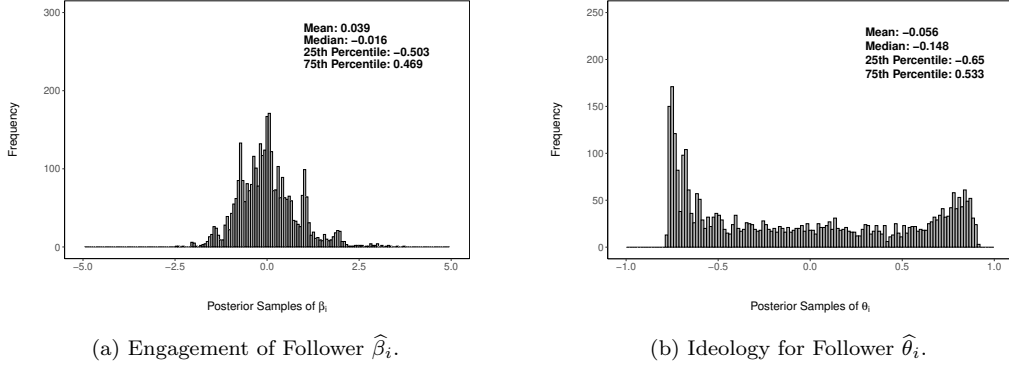


Figure 4. Histograms and summary statistics for the estimates of (a) engagements $\hat{\beta}_i$ and (b) ideal points $\hat{\theta}_i$ for the 1285 followers in the simulated Twitter data.

threshold, indicated by the red horizontal line, which is the recommended standard used in MC-MC estimation [4].

2.4.2. Prediction Diagnostics

Figure 6 is a heat map of the estimates $\hat{p}_{ij}$ of the probabilities $p_{ij} = \Pr(\text{follower } j \text{ follows influencer } i)$. A quick visual inspection shows that the structure of the $\hat{p}_{ij}$'s in Figure 6 closely mimics that of the y_{ij} 's in Figure 1. This indicates adequate model fit and prediction diagnostics.

A common approach to improve prediction is to find the threshold value $\tau_{\hat{p}}$ that maximizes [5]'s J -statistic ([6,7])

We generate a confusion matrix using an optimal threshold of $\tau_{\hat{p}} = 0.246$. The optimal threshold is generated by maximizing [5]'s J -statistic, and this approach is commonly used to improve prediction ([6,7]). The diagnostic results are presented in Figure 7 (a). Note that the overall accuracy of this model is $0.206 + 0.655 = .861$, which indicates the model is 86.1% accurate. Figure 7 (b) is the receiver operating characteristic (ROC) curve, which has as desirable shape. More details about the ROC curve can be found in [1].

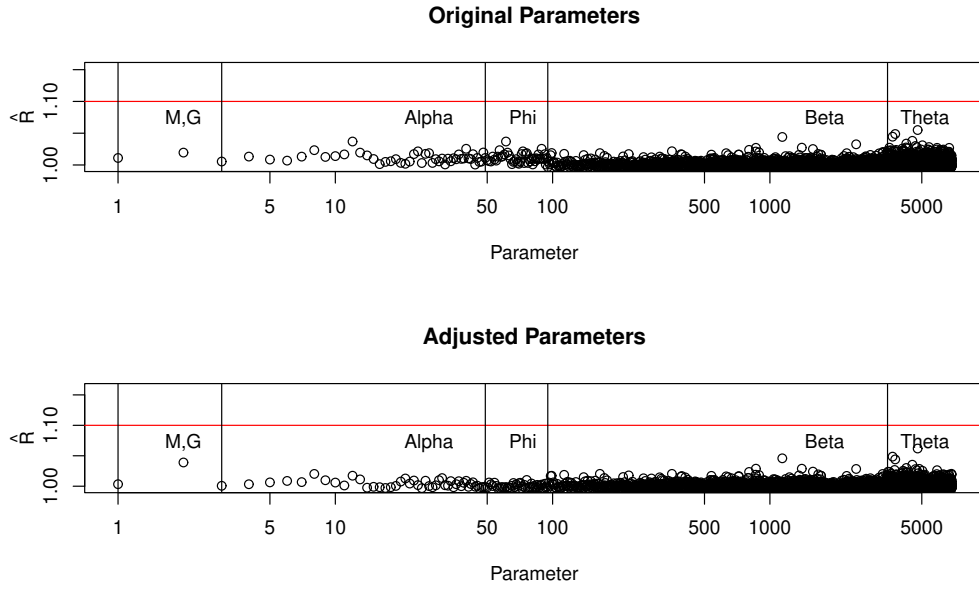


Figure 5. $\hat{R}$ diagnostic plots for the original parameters (top) and adjusted parameters (bottom) for the simulated Twitter data.

References

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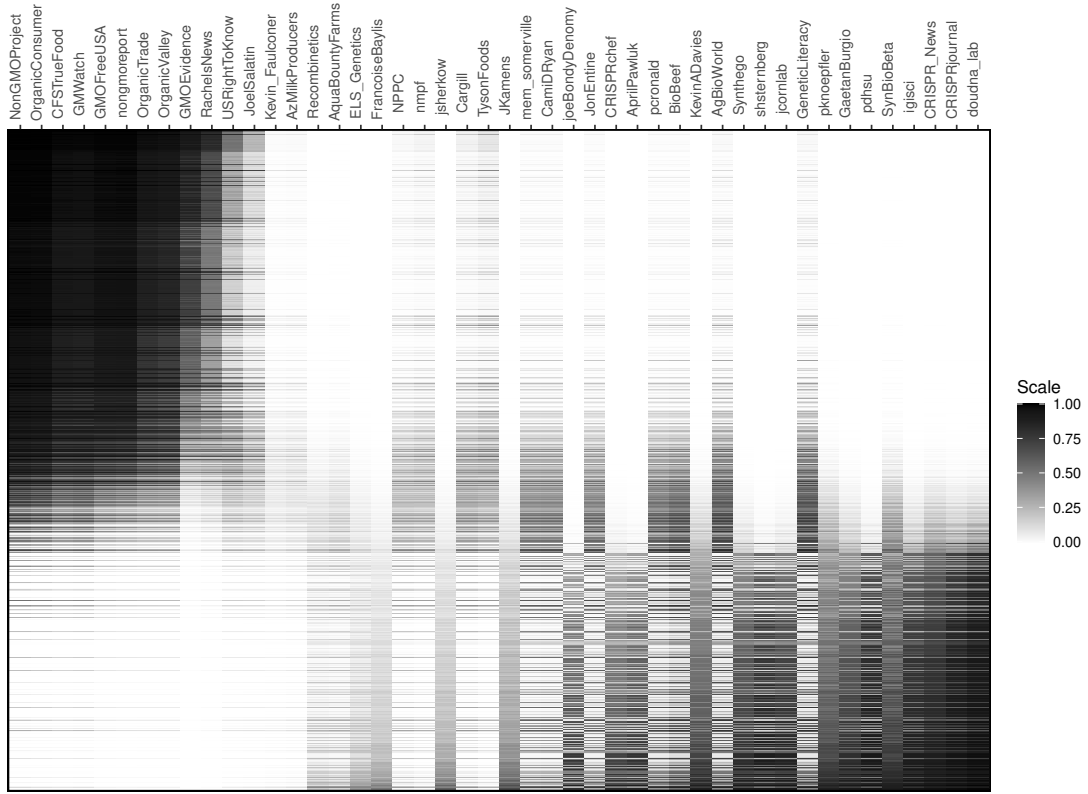


Figure 6. Heat map of estimates $\hat{p}_{ij}$ of the probabilities $p_{ij} = \Pr(\text{follower } i \text{ follows expert } j)$.

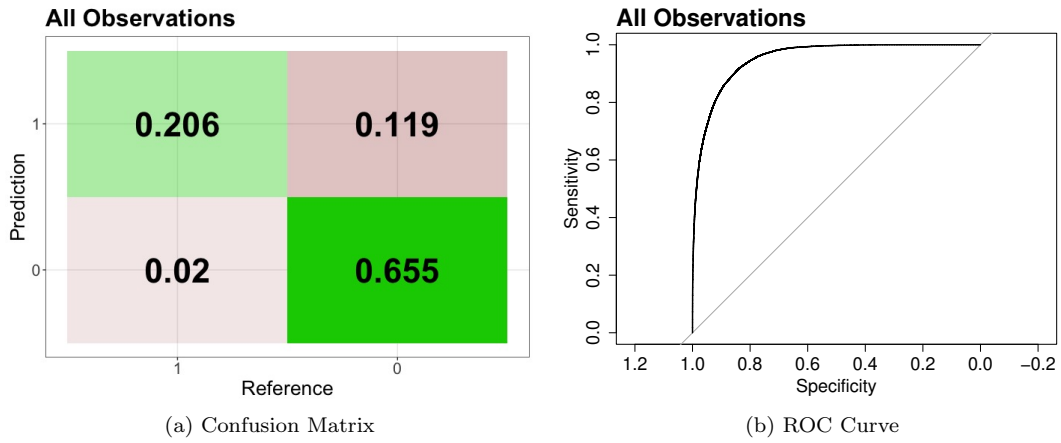


Figure 7. The confusion matrix of classification proportions (a) with prediction threshold 0.50 for $\hat{p}_{ij}$ and a plot of the ROC curve (b) for the simulated Twitter data.